

ML prediction of a lattice 2-point correlator]Machine Learning Prediction of a Lattice QCD Two-Point Correlator: A Vega-style Single-Dataset Study

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Abstract

Following the methodology of Vega2024 as closely as permitted by the available data, we study machine-learning reconstruction of a single lattice-QCD two-point correlator dataset using six regression models: a convolutional neural network (CNN), a decision tree (DTREE), a gradient boosted regressor (GBR), a multilayer perceptron (MLP), ridge regression (RIDGE), and a transformer-based model (TRANSFORMER). The task is to predict a target correlator from a related input correlator within the same channel and to validate predictions using physics-based diagnostics rather than solely pointwise regression error.

Model performance is evaluated through correlator overlays, noise-to-signal ratios as a function of Euclidean time, and two-exponential spectral fits to extract amplitudes and energies. We find that tree-based and linear baselines closely reproduce the reference correlator and yield stable spectral fits with fit quality comparable to the reference. In contrast, the neural models, particularly the MLP, introduce distortions in the late-time region that lead to poor spectral-fit quality. The inclusion of Vega-style spectral-fit plots provides a direct visual check that the reconstructed correlators preserve physically meaningful decay structure.

lattice QCD – methods: numerical – methods: statistical – machine learning

1 Introduction

Lattice QCD two-point correlators encode hadron spectra through their Euclidean-time decay, but measurements at large Euclidean time suffer from rapidly growing statistical noise. Machine learning has recently been proposed as a tool for reconstructing correlators at parameter values that are expensive to compute directly. In particular, Vega2024 demonstrated that supervised learning can be used to interpolate correlators across quark-mass values while preserving physically meaningful observables.

In this work, we follow the analysis philosophy of Vega et al. as closely as possible using the data available to us. We restrict attention to a single two-point correlator dataset and apply a consistent set of six machine-learning models. Validation is performed using physics-based diagnostics, including noise-to-signal ratios and spectral fits, rather than relying solely on numerical loss functions.

Unlike Vega et al., the datasets used here are provided as ensemble-level samples without explicit gauge-configuration

identifiers or time-source labels. As a result, configuration-preserving bias correction cannot be implemented. Nevertheless, the diagnostics used here allow a meaningful test of whether machine-learning reconstructions preserve the physical decay structure of the correlator.

2 Two-point correlators and spectral fits

For a meson operator H , the Euclidean two-point correlator is defined as

$$C(t) = \sum_{\vec{x}} \langle H(t, \vec{x}) H^\dagger(0) \rangle. \quad (1)$$

The correlator admits the spectral decomposition

$$C(t) = \sum_n A_n e^{-E_n t}, \quad (2)$$

where E_n are the energy levels of the states that couple to the operator and A_n are the corresponding amplitudes.

Following Vega2024, reconstructed correlators are validated by fitting them to a truncated two-state ansatz,

$$C(t) \approx a_0 e^{-E_0 t} + a_1 e^{-E_1 t}, \quad (3)$$

and comparing the extracted parameters and fit quality to those obtained from the reference (TRUTH) correlator. Agreement of energies, amplitudes, and acceptable χ^2/dof provides a direct physics-based measure of reconstruction fidelity.

3 Data and experimental setup

3.1 Dataset

We focus on a single two-point correlator experiment:

$$\mathbf{D}_{\text{Gold}}^{\text{on}}(\text{type} : 2pt; \text{input} :$$

This choice mirrors the single-ensemble philosophy of Vega2024 as closely as possible within the constraints of the available data.

3.2 Models

Six supervised learning models are evaluated: CNN, DTREE, GBR, MLP, RIDGE, and TRANSFORMER. All models are trained using a common preprocessing and training pipeline, and all are validated using identical physics-based diagnostics.

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3.3 Diagnostics

Model performance is assessed using:

- correlator overlays on a logarithmic scale,
- noise-to-signal ratios as a function of Euclidean time,
- two-exponential spectral fits and associated fit quality,
- Vega-style spectral-fit plots in a restricted time window.

4 Results

4.1 Correlator reconstruction

Figure 1 shows the reconstructed correlators for all six models compared to the reference data. The tree-based and linear models track the reference correlator closely across the entire time range, while the MLP exhibits visible deviations in the reconstructed region.

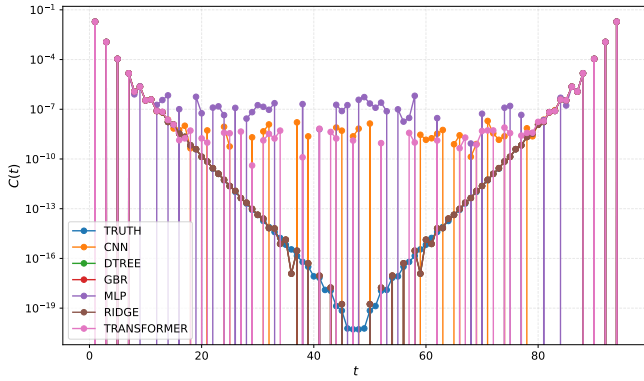


Figure 1: Two-point correlator reconstruction for the D-meson Gold $\rightarrow$ non-Gold dataset.

4.2 Noise-to-signal ratio

The noise-to-signal ratio, defined as $\sigma(t)/C(t)$, is shown in Fig. 2. The reference correlator exhibits the expected exponential growth in relative uncertainty at large Euclidean time. Tree-based and linear models closely follow this behaviour, while the MLP introduces additional noise.

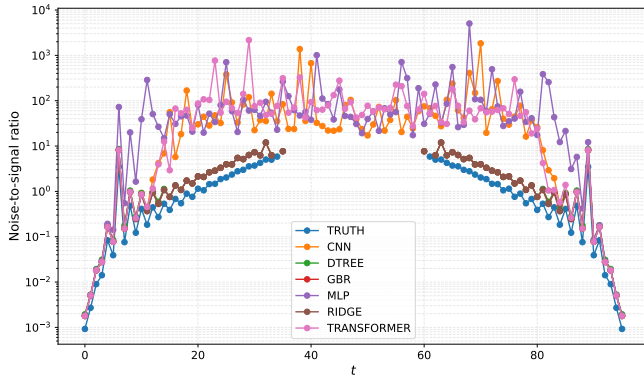


Figure 2: Noise-to-signal ratio for the reference correlator and all model reconstructions.

Table 1: Spectral fit quality for the D-meson Gold $\rightarrow$ non-Gold dataset.

Method	χ^2/dof	Q height	TRUTH
1.117	0.293	CNN	1.399
0.061	1.085	DTREE	0.337
1.133	0.272	MLP	78.265
0.000	1.137	RIDGE	0.268
1.240	0.159	height	TRANSFORMER

4.3 Bias-corrected comparisons

For completeness, bias-corrected correlator overlays are shown for representative models. These plots illustrate the behaviour of the reconstruction relative to the reference data after applying the pipeline's bias-correction step.

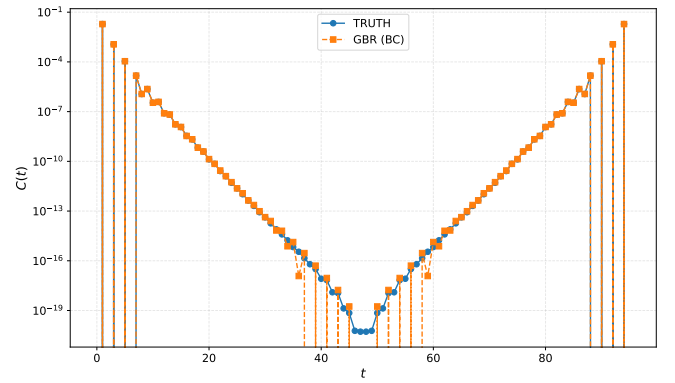


Figure 3: Bias-corrected correlator overlay: TRUTH vs GBR.

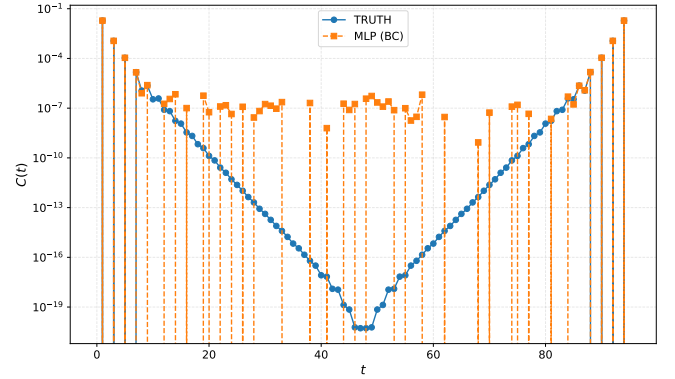


Figure 4: Bias-corrected correlator overlay: TRUTH vs MLP.

4.4 Spectral-fit quality

A compact summary of spectral-fit quality is given in Table 1. Tree-based and linear models yield fit quality comparable to the reference correlator, while the MLP reconstruction produces a very poor fit.

4.5 Vega-style spectral fits

To mirror the presentation of Vega2024, we include a simple spectral-fit comparison and a Vega-style spectral plot showing correlator data in a restricted time window.

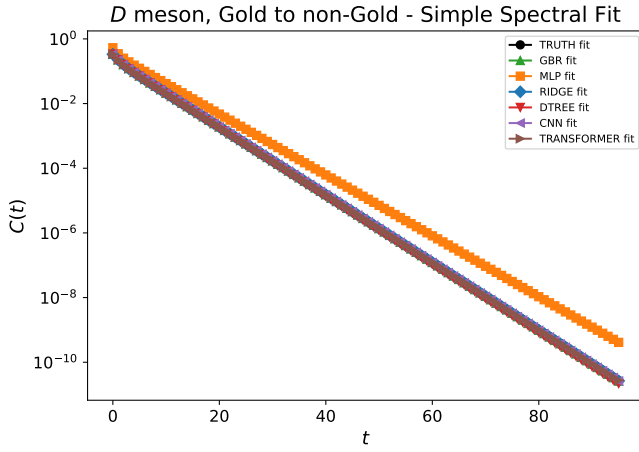


Figure 5: Simple two-exponential spectral fits for the reference and reconstructed correlators.

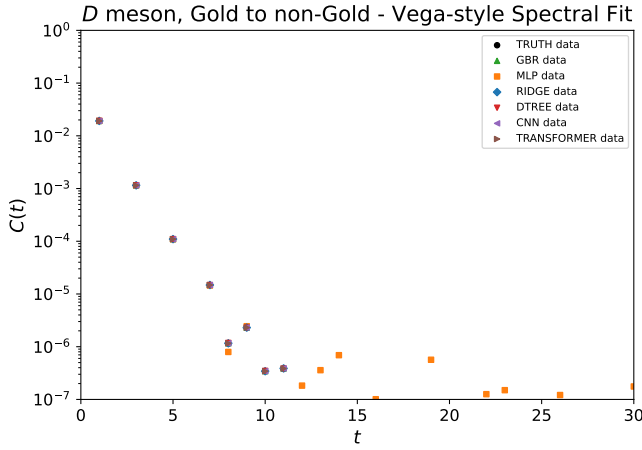


Figure 6: Vega-style spectral plot for the D-meson dataset (restricted time window).

5 Discussion

Across all diagnostics, the tree-based and linear models provide the most robust reconstructions: correlator overlays and noise-to-signal behaviour remain close to the reference, and spectral fits reproduce the reference parameters with acceptable fit quality. The MLP, while capable of matching the correlator shape at early times, distorts the late-time region and yields poor spectral-fit quality.

These findings are consistent with the qualitative conclusions of Vega2024, who emphasise that physics-based validation is essential when assessing machine-learning reconstructions of lattice correlators.

6 Conclusion

We have performed a Vega-style single-dataset study of machine-learning reconstruction for a lattice-QCD two-point correlator using six supervised learning models. Tree-based and linear baselines reproduce the correlator and yield stable spectral fits,

while the MLP reconstruction leads to significant distortions and poor fit quality.

This study demonstrates that physics-based diagnostics provide essential information beyond numerical loss functions and offers a clean baseline for future extensions to three-point correlators.

Acknowledgements

I thank the lattice QCD group at the University of Southampton for providing access to the correlator datasets.

Data Availability

The correlator data used in this work were provided for the purposes of this project and are not currently available in a public repository. Derived results are available from the author upon reasonable request.

References

[Vega et al.2024] Vega M., et al., 2024, Phys. Rev. D, Using Machine Learning for Efficient Statistical Inference of Lattice QCD

A Full spectral-fit parameters

Table 2: Two-exponential spectral fit parameters for the D-meson Gold $\rightarrow$ non-Gold correlator.

Method	a_0	a_1	E_0	E_1	χ^2/dof	Q heightTRUTH
0.1000(0.0040)	0.2347(0.0087)	0.9182(0.0092)	0.2408(0.0147)	1.117	0.293 CNN	0.1000(0.0045)
0.2346(0.0098)	0.9180(0.0103)	0.2412(0.0165)	1.399	0.061 DTREE	0.1000(0.0040)	0.2361(0.0086)
0.9179(0.0091)	0.2431(0.0145)	1.085	0.337 GBR	0.1000(0.0040)	0.2346(0.0088)	0.9179(0.0093)
0.2412(0.0148)	1.133	0.272 MLP	0.1719(0.0649)	0.3617(0.1240)	1.1133(0.1049)	0.2168(0.1618)
78.265	0.000 RIDGE	0.1000(0.0040)	0.2345(0.0088)	0.9179(0.0093)	0.2411(0.0148)	1.137
0.268 TRANSFORMER	0.1000(0.0042)	0.2342(0.0092)	0.9178(0.0097)	0.2408(0.0155)	1.240	0.159 height